Ram Calculus and Planar Halving Lines

Exact Sweep Ledgers, Sharp Local Bounds,
Conditional Exponent Transfer, and a Source Audit

Technical Report TR-M334-Ram-Halving

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Abstract

We examine the September 25, 2026 preprint *A power saving for planar halving lines*, pinned to the public commit specified in this report, against the confidential Ram Calculus corpus, especially the complete revised TR-item-214-astra. The source claims absolute $\varepsilon > 0, C < \infty, n_0$ such that every even $n \geq n_0$ point set with no three collinear determines at most $Cn^{4/3-\varepsilon}$ unordered halving pairs. Its claim is nonquantitative and is not treated here as globally verified.

We prove exact half-turn and full-turn accounting, active-subset rank restriction, weighted signed telescoping, sharp variation constants, and explicit conditional concentration and recurrence transfer theorems. The source's finite three-strip boundary estimate is strengthened from $2C_B$ to the sharp C_B by separate exit and entry accounting. At a supplied valid scale the recurrence also yields a sharper conditional exponent than the source's slackened induction. These are local algebraic or finite geometric improvements, not a proof of uniform activity packing.

Every substantive source section is read and mapped to its mathematical role, including rank limits, differentiability, three-foliation zooms, spatial measure selection, boundary activity, and the final recurrence. The missing input in an independent Ram proof remains a uniform realizability-sensitive geometric estimate. Deterministic rational checks, source hashes, build diagnostics, and a final output manifest make the scope of the investigation reproducible without exporting private material.

Reading rule. *Proved here* applies only to the explicitly stated local results. *Source claim* and *conditional* are not synonyms for independently established asymptotic geometry.

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1 Pinned Claim and Verification Boundary

Let $P \subset \mathbb{R}^2$ have even cardinality n, with no three collinear. An unordered pair $\{p, q\}$ is halving when the line through it has $(n-2)/2$ other points in each open half-plane. Write $h(P)$ for the number of such pairs. No two different pairs determine the same line under this hypothesis.

The exact source assertion is

$$\exists \varepsilon > 0 \; \exists C < \infty \; \exists n_0 \; \forall n \in 2\mathbb{N}, \; n \geq n_0 \; \forall P, \quad h(P) \leq C n^{4/3-\varepsilon}. \tag{1.1}$$

The constants are absolute, not configuration-dependent. The source supplies no numerical ε, C, n_0. Its stronger intermediate claim bounds switches uniformly over *all* ranks and all sizes. This stronger rank quantifier is necessary when restricting to active points inside a block.

The source is OpenAI's preprint [1], at commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The collection README explicitly describes results at different verification stages and warns that unformalized results can have issues [2]. We have not checked any Lean artifacts or independently certified the entire compactness argument. A complete source read is not a global proof certificate.

Three separate outputs. First, this report establishes elementary finite lemmas with full proofs. Second, it proves transfer theorems whose hypotheses are stated explicitly. Third, it audits the architecture of the source's claimed geometric proof without importing that claim as a theorem of Ram Calculus. The report does not assert a counterexample to (1.1), and does not assert that it is established.

Locality and confidentiality. The source originals and private corpus originals are unchanged. Private files were read locally; public browsing used only public titles, authors, mathematical topics, and public URLs. The new report, checks, and evidence live in the assigned M334 directory; export artifacts live under the report's named build directory. No shared catalogue, progress file, queue, or helper was edited.

Notation. S_j counts rank-j switches in one generic half-turn; $f_k(P)$ counts distinct strictly separable k-subsets. N is used for a general finite sweep size. A block's activity counts distinct points, not events and not only its endpoint symmetric difference. For source crossing arguments, genericity additionally excludes a proper crossing of two determined segments on a third segment. Constants carrying a subscript G may depend on a selected limiting set; they must not be silently promoted to uniform finite constants.

2 Primary Literature and Source Status

The public literature check was conducted on October 7, 2026. Dey's published paper proves the planar upper bound $O(n(k+1)^{1/3})$ [3]; at middle rank this is $O(n^{4/3})$. Tóth's construction gives the lower scale $n\exp(\Omega(\sqrt{\log k}))$ [4]. These statements describe established reference bounds, not an exhaustive certification of every subsequent announcement.

Alonso, López, and Rodrigo's 2024 article improves lower-order terms and finite-size upper bounds using squared degrees of halving graphs [5]. Its contribution does not replace 4/3 by a smaller fixed exponent. The publisher endpoint was rate-limited during this check; an institutional deposited published version was accessible. The 2025 odd-cardinality construction paper [6] concerns different parity conventions and finite examples. It is not a substitute for the uniform even-n statement (1.1).

The source invokes shallow cuttings through Agarwal, Aronov, Chan, and Sharir's author manuscript [7]. We directly inspected the statement of Lemma 3.2 and its surrounding reduction. For planar lines, choosing $r = n/(k+1)$ yields $O(n/(k+1))$ cells of conflict size $O(k+1)$, with rank offsets changing between cells. The exact local rank must therefore be retained.

The three-foliation discussion is related to the affine-rigidity literature of Artstein-Avidan and Slomka [8]; the browsed author abstract concerns maps preserving specified families of lines. That background does not automatically establish the source's stronger sequential noncollapse lemma with moving base points.

Current-status discipline. The pinned M334 is a new source claim beyond the classical benchmark. The browsed collection README does not certify this particular proof. No claim is made here that an internet search establishes absence of other developments, peer-review acceptance, or completeness of the formalization catalogue. The appropriate comparison is

published reference bounds versus new pinned preprint claim under audit.

All locally developed claims below are independent of whether the preprint is ultimately accepted.

Attribution. The sweep ledger and sharp unweighted variation formula are already present in the revised private TR-item-214 [9]. The present report supplies their explicit weighted and error-aware deployment, an exact recurrence sharpening, and a sharp finite boundary constant. We do not claim historical priority for telescoping, discrete Fourier inversion, crossing inequalities, or recurrence induction.

3 Complete Source and Corpus Correspondence

All eight source sections, the main file, and the bibliography were read completely. The following is a proof-dependency map, not merely a table of shared terminology.

Source file	Mathematical task	Local correspondence and boundary
`introduction.tex`	Absolute saving; all ranks; k-sets	Exact quantifiers in Section 1; no adopted global claim.
`finite.tex`	Activity, batches, exchange graphs, crossings	Sections 4, 6, 13; TR214 event ledger and realizability boundary.
`rank-limits.tex`	Compact labels; endpoint, vertex, crossing measures	Section 18; normalization and absolute-continuity audit.
`differentiability.tex`	Monotone charts, critical images, transverse directions	Section 19; analytical inputs are not spectral identities.
`zoom.tex`	Sequential three-foliation rigidity and radial comparisons	Section 19; fixed-radius quantifiers retained.
`spatial.tex`	Positive edge mass with controlled vertex density	Section 20; measure types and graph limits kept distinct.
`boundary.tex`	Three strips, tokens, limiting contradiction	Sections 21–23; sharp finite C_B improvement proved here.
`recurrence.tex`	Fixed-scale recursion; shallow cuttings	Sections 15, 16, 24; conditional endpoint exponent and exact counting.

The private full TR214 was read through its revisions and bibliography. Its conditional concentration theorem gives the baseline transfer; its revision gives the optimal local variation coefficient and exact full-turn count. The earlier, weaker wording is not substituted for the revised assertions.

Actual additional private passages were read, rather than relying on routing maps: TR9, lines 1–240, for the distinction between a standard DFT and a general shift-variant kernel; TR190, lines 1–260, for the spectral operator and inverse tiers; and TR165, lines 183–220, 265–274, and 365–384, for residual/error and scope discussions [10, 12, 11]. We do not claim full audits of these three larger reports. Exact absolute paths, file sizes, line counts, and hashes are recorded in `evidence/input-hashes.json`.

4 Exact Finite Sweep Ledger

Assume first that the x_i are distinct, no three points are collinear, and no two determined slopes coincide. Order points by

$$g_i(s) = y_i - sx_i, \qquad -\infty < s < \infty.$$

At the slope $s_{ij} = (y_j - y_i)/(x_j - x_i)$ exactly two projections coincide. They occupy adjacent positions: a point strictly between their equal values is impossible, and a third equal value is excluded. Let $S_j(P)$ count swaps between positions j and $j+1$.

Theorem 4.1 (Half-turn ledger)**.** *For $N \geq 2$,*

$$\sum_{j=1}^{N-1} S_j(P) = \binom{N}{2}. \tag{4.1}$$

When $N = 2m$, $h(P) = S_m(P)$.

Proof. Each affine difference $g_i - g_j$ has exactly one real zero. Every unordered pair therefore contributes exactly once. At a rank-j event, $j-1$ points have projection strictly below the pair and $N-j-1$ strictly above it. These are exactly the counts in the two open half-planes of the pair's line. They are equal precisely when $j = m$. □

As s increases, the newly marked point at any fixed rank has larger x-coordinate than the lost point. Indeed, the point with larger x has the more negative derivative in $g_i(s)$. Thus

$$\Phi_k(s) = \sum_{i \in M_k(s)} x_i \tag{4.2}$$

strictly increases at each rank-k switch. A half-turn cannot return to a previous marked set, although individual points can return.

Removing extra genericity. For at least three points with no three collinear, finitely many nonzero triple determinants have a positive minimum absolute margin. A sufficiently small perturbation preserves their signs and hence all side counts. The additional nongeneric conditions are contained in finitely many proper algebraic zero sets. A perturbation avoiding them preserves all halving pairs. With two points, preserve distinctness and choose distinct first coordinates. Exact computations may instead resolve simultaneous disjoint pair events, but must never order a triple tie as though it were generic.

What the identity does not say. The mean of the $N-1$ coordinates is $N/2$. This mean imposes no $N^{4/3}$ bound on a designated coordinate of an unrestricted nonnegative vector. A vector placing all $\binom{N}{2}$ at one coordinate satisfies the identity. It need not be a realizable sweep spectrum. The absent restriction is geometric realizability, not invertibility of a transform.

5 Full-Turn Counting Without Extra States

Let $u(\theta) = (\cos\theta, \sin\theta)$ and choose a regular starting angle. A half-turn gives one reduced sequence from an initial order to its reverse. Denote its rank counts by $S_j(P;\theta_0)$. For $1 \le k < N$, let $f_k(P)$ be the number of distinct k-subsets strictly separated from their complements by an affine line.

Theorem 5.1 (Cyclic state formula)**.**

$$f_k(P) = S_k(P;\theta_0) + S_{N-k}(P;\theta_0). \tag{5.1}$$

In particular, $f_{N/2}(P) = 2h(P)$ *for even* N*.*

Proof. A fixed k-subset Q is a projection prefix exactly when

$$\langle p_j - p_i, u\rangle > 0 \qquad (i \in Q,\ j \notin Q).$$

These strict homogeneous inequalities define an open convex cone. Because Q is nonempty and proper, one of the inequalities confines it to an open half-plane. Its intersection with the unit circle, if nonempty, is one connected arc. Hence each separable subset occurs once in the cyclic full-turn state sequence. Every transition is a rank-k swap; the number of states equals the number of transitions on a cycle. Negating the normal reverses ranks, so the second half-turn contributes the first half's rank-$(N-k)$ count. □

A linear half-turn lists $1 + S_k$ states. Adding two such lists adds two redundant endpoint states. This explains why the source's $S_k + S_{N-k} + 2$ upper bound is valid but not exact. Equation (5.1) removes the additive slack; it is not an exponent improvement.

Corollary 5.2 (Determinant-only count)**.**

$$f_k(P) = \#\{(i,j) : i \ne j,\ \#\{\ell \ne i,j : \det(p_j - p_i, p_\ell - p_i) > 0\} = k-1\}.$$

Proof. Each oriented pair with the stated side count gives one oriented rank-k event. Rotating the normal slightly past it chooses the entering state. Conversely every cyclic transition has such an oriented pair. Reversing an oriented pair interchanges the two side counts, as required by the two half-turns. □

For rational points all these tests use exact arithmetic. At even middle rank each unordered halving pair is counted twice by the oriented formula and once by $S_{N/2}$.

6 Activity and Active-Subset Restriction

For consecutive sweep states $M_0, \dots, M_t$ of fixed size k, define

$$A = \bigcup_{i=1}^{t} (M_{i-1} \triangle M_i), \qquad a_{\mathrm{act}} = |A|.$$

The endpoint distance $d(M_0, M_t) = |M_0 \setminus M_t|$ is a different quantity. Points can enter and leave between endpoints without contributing to their symmetric difference.

Lemma 6.1 (Exact rank restriction)**.** *Put $B = M_0 \setminus A$ and $r = k - |B|$. At every state in the block, $M_i \cap A$ consists of the r lowest projected points of A. The rank-k switches of the original block are exactly the rank-r switches of the restricted configuration on that slope interval.*

Proof. Every point outside A has constant membership, so precisely the points of B remain marked outside A. Thus $|M_i \cap A| = r$. Every marked point has smaller projection than every unmarked point at a regular state. Restricting this comparison to A proves the prefix assertion. Both members of an actual rank-k exchange lie in A, and the restricted prefix changes there. Conversely a restricted rank-r exchange changes $M_i \cap A$, so the fixed set B implies a change of the original prefix. □

Lemma 6.2 (Greedy activity overshoot)**.** *For an integer threshold $b \geq 1$, stop a block the first time its activity reaches b. Its activity is at most $b + 1$. An unfinished remainder has activity less than b.*

Proof. One event changes two memberships, and a nonswitch changes none. The final event introduces at most two previously inactive points. Immediately before it activity is at most $b - 1$. This proves the upper bound; the stopping rule proves the other statements. □

The local rank r need not be half the active-set size and can vary from block to block. A middle-rank bound alone does not close the recurrence. Neither is it necessary to assume monotonicity of $F(n)$ in n: induction can be applied separately to the actual active sizes.

Corpus connection. TR214's event ledger survives this restriction as an exact finite object. A graph-coarsening matrix from TR190 generally does not preserve the same incidence data or local rank. Active restriction is justified by the membership argument above, not by the existence of an inverse on a retained spectral subspace.

7 Weighted Signed Telescoping

Let $S_a, \dots, S_b$ be real, $a \le k \le b$, and let $w_j \ge 0$ with $\sum_{j=a}^{b} w_j = 1$. Define $\Delta_j = S_{j+1} - S_j$ and

$$L_j = \sum_{i=a}^{j} w_i, \qquad R_j = \sum_{i=j+1}^{b} w_i.$$

No geometry is required for the following statement.

Theorem 7.1 (Weighted exact residual)**.**

$$S_k - \sum_{i=a}^{b} w_i S_i = \sum_{j=a}^{k-1} L_j \Delta_j - \sum_{j=k}^{b-1} R_j \Delta_j. \tag{7.1}$$

Consequently

$$S_k \le \sum_{i} w_i S_i + \sum_{j<k} L_j (\Delta_j)_+ + \sum_{j \ge k} R_j (-\Delta_j)_+, \tag{7.2}$$

and

$$S_k \le \sum_{i} w_i S_i + c_w \sum_{j=a}^{b-1} |\Delta_j|, \quad c_w = \max\left\{ \sum_{i<k} w_i, \sum_{i>k} w_i \right\}. \tag{7.3}$$

Proof. For $i < k$, write $S_k - S_i = \sum_{j=i}^{k-1} \Delta_j$. For $i > k$, write $S_k - S_i = -\sum_{j=k}^{i-1} \Delta_j$. Multiply by w_i and interchange the finite sums. This gives (7.1). Removing negative terms gives (7.2). The largest L_j on the left is L_{k-1}, and the largest R_j on the right is R_k. Their maximum is c_w, proving (7.3). □

Exact core and residual. The weighted mean is an explicitly chosen core. The right side of (7.1) is its signed residual. This is a rigorous finite realization of the organizational principle discussed in TR165. The decomposition itself does not make the residual small. In particular, no trained residual or empirical sample substitutes for a bound on every realizable configuration.

Coefficient design. If separate geometric bounds are known for weighted mass and each one-sided difference, (7.2) gives a linear expression in the weights. Minimizing that expression over the simplex of weights is a finite linear optimization problem. Putting all weight at k makes $c_w = 0$, but also makes the mean exactly S_k: this is a tautology, not an estimate.

8 Sharpness and Error-Aware Certificates

For uniform weights on $W = [a, b] \cap \mathbb{Z}$ with $m = b - a + 1$, Theorem 7.1 gives

$$S_k - \frac{1}{m}\sum_{i=a}^{b} S_i = \sum_{j=a}^{k-1} \frac{j-a+1}{m}\Delta_j - \sum_{j=k}^{b-1} \frac{b-j}{m}\Delta_j, \tag{8.1}$$

$$S_k \le \frac{1}{m}\sum_{i=a}^{b} S_i + c_W \,\mathrm{TV}_W(S), \tag{8.2}$$

$$c_W = \frac{\max(k-a, b-k)}{m}, \quad \mathrm{TV}_W(S) = \sum_{j=a}^{b-1} |\Delta_j|. \tag{8.3}$$

These are the revised TR214 formulas, derived here for completeness.

Proposition 8.1 (Optimal coefficient)**.** *The coefficient c_w in* (7.3) *is optimal over all nonnegative sequences. For $m = 1$ it is zero. In a symmetric, untruncated radius-r window, $c_W = r/(2r+1) < 1/2$.*

Proof. If the left mass is the larger, take $S_i = 0$ for $i < k$ and $S_i = 1$ for $i \ge k$. Its total variation is one and the left side of (7.1) is exactly $\sum_{i<k} w_i$. If the right mass is larger, take $S_i = 1$ for $i \le k$ and $S_i = 0$ for $i > k$. This attains the other coefficient. Empty sides and the singleton case are immediate. □

The extremizing step vectors need not be realizable point spectra. Thus sharpness for arbitrary sequences does not preclude stronger geometric inequalities for actual halving configurations.

Proposition 8.2 (Certified perturbation of a ledger)**.** *Suppose $|S_i - \widetilde{S}_i| \le e_i$ on W, and define*

$$\widetilde{M} = \sum_i w_i \widetilde{S}_i, \quad E_w = \sum_i w_i e_i, \quad E_\Delta = \sum_{j=a}^{b-1} (e_j + e_{j+1}).$$

Then

$$S_k \le \widetilde{M} + E_w + c_w(\mathrm{TV}_W(\widetilde{S}) + E_\Delta).$$

Proof. Bound the true weighted mean by $\widetilde{M} + E_w$. The triangle inequality gives $|\Delta_j| \le |\widetilde{S}_{j+1} - \widetilde{S}_j| + e_j + e_{j+1}$. Insert these two estimates in (7.3). □

When all $e_i \le e$, the additional certified error is at most $e + 2c_w(m-1)e$. Small transform reconstruction errors therefore require a quantified margin; they cannot be discarded at growing window size.

9 Joint Mass and Variation: Exact Optimum

The independent bounded audit [13] supplies a useful strengthening that also exploits nonnegativity. We reproduce its proof and attainment, keeping its scope separate from realizability.

Theorem 9.1 (Sharp joint bound)**.** *Let $S_a, \dots, S_b \ge 0$, $\sum S_i \le M$ and $\mathrm{TV}_W(S) \le D$, where $M, D \ge 0$. Put $L = \max(k-a, b-k)$, $R = \min(k-a, b-k)$, and $m = L + R + 1$. Then the exact unrestricted optimum is*

$$S_k \le \min\left\{M, \frac{M + LD}{m}, \frac{M + RD}{2R+1}\right\}. \tag{9.1}$$

Every M, D admits a nonnegative real sequence attaining the bound.

Proof. Write $x = S_k$, and let v_L, v_R be variations on the longer and shorter sides. Each side entry is at least $(x - v_{\text{side}})_+$. Thus the mass is at least $mx - L\min(x, v_L) - R\min(x, v_R)$. The subtracted term is at most LD. It is also at most $(L-R)x + RD$, since $L\min(x, v_L) + R\min(x, v_R) = (L-R)\min(x, v_L) + R(\min(x, v_L) + \min(x, v_R))$. Together with mass at least x, these give the three bounds.

For attainment use a constant value on each side. If $M \ge (R+1)D$, choose $x = (M + LD)/m$, longer-side value $x - D$, and shorter-side value x. If $D/2 \le M \le (R+1)D$, choose $x = (M + RD)/(2R+1)$, longer-side value zero, and shorter-side value $2x - D$. If $M \le D/2$, choose $x = M$ and both sides zero. Each construction has mass M, nonnegative entries, and variation at most D. Comparing the three expressions verifies which is minimal on each range. Empty sides simply omit their entries and jumps; for $m = 1$ the answer is M. □

For $L = 4, R = 1, M = D = 1$, the variation-only estimate gives $5/6$, while (9.1) gives $2/3$, attained by $(0, 0, 0, 0, 2/3, 1/3)$. These vectors are not claimed realizable. If uniform geometric estimates give mass $M \le C_1 mT$ and variation $D \le C_2 T$, then

$$S_k \le T \min\left\{C_1 m,\ C_1 + \frac{L}{m} C_2,\ \frac{C_1 m + RC_2}{2R+1}\right\}.$$

Here $T = n(k+1)^\alpha$ may be used, but the geometric bounds still must be proved uniformly. Different half-turns may use different windows and must use their respective constants.

10 Conditional Concentration Transfer

Hypothesis 10.1 (Uniform local concentration)**.** Fix $0 \le \alpha < 1/3$ and $C_1, C_2 \ge 0$ independent of all configurations. For every realizable n-point spectrum, every regular starting angle, and every $1 \le k \le n/2$, suppose there is an interval $W = [a, b]$ containing k, with $m = |W|$, such that

$$\sum_{j \in W} S_j \le C_1 mn(k+1)^\alpha, \qquad \mathrm{TV}_W(S) \le C_2 n(k+1)^\alpha.$$

Suppose also that $c_W \le c_*$ for a uniform specified $c_* < 1$.

Theorem 10.2 (Explicit transfer)**.** *Under Hypothesis 10.1, set $K = C_1 + c_* C_2$. Then*

$$S_k(P; \theta_0) \le Kn(k+1)^\alpha, \tag{10.1}$$

$$f_k(P) \le 2Kn(k+1)^\alpha, \tag{10.2}$$

$$h(P) \le Kn^{1+\alpha} \qquad (n \text{ even},\ n \ge 2). \tag{10.3}$$

The halving exponent saving is $\varepsilon = 1/3 - \alpha$.

Proof. Insert the two bounds in (8.2). For the second half-turn, apply the same hypothesis at starting angle $\theta_0 + \pi$ and rank k. Its count is $S_{n-k}(P; \theta_0)$. Add the two estimates and use Theorem 5.1. At middle rank use $h = S_{n/2}$ and $(n/2+1)^\alpha \le n^\alpha$. □

This strengthens the constant $C_1 + C_2$ in the earlier version of TR214, but its exponent is entirely supplied by the hypothesis. Untruncated symmetric windows permit $c_* = 1/2$, with a strictly smaller finite-window coefficient whenever a uniform radius is fixed. Truncated edge windows must use their actual asymmetric coefficients.

Residual variant. It suffices instead to bound the signed residual in (8.1) by $C_2 n(k+1)^\alpha$; total variation is a convenient sufficient bound, not a necessary one. Oppositely signed increments can cancel. A useful refinement would prove such cancellation uniformly for geometrically realizable spectra.

Quantifier warning. The interval may depend on the configuration, rank, and starting angle; the constants may not. Checking this condition for finitely many configurations does not establish it. Permitting radius zero reduces its mass condition to the desired conclusion itself, which is logically valid but gives no new method. This report establishes neither the hypothesis with $\alpha < 1/3$ nor its signed variant.

11 Transform Coordinates and Their Limits

Extend S_j by zero outside $1 \le j \le n-1$, and define

$$\widehat{S}(\xi) = \sum_{j=1}^{n-1} S_j e^{-ij\xi}, \qquad S_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} \widehat{S}(\xi) e^{ik\xi} \, \mathrm{d}\xi.$$

Both formulas are ordinary finite Fourier analysis, interpreted here as the discrete specialization used in TR214. Orthogonality of the exponentials proves inversion without a convergence issue.

Writing $z = e^{-i\xi}$ and multiplying the polynomial $\sum S_j z^j$ by $1 - z$ gives the exact endpoint identity

$$(1 - e^{-i\xi})\widehat{S}(\xi) = S_1 e^{-i\xi} - S_{n-1} e^{-in\xi} + \sum_{j=1}^{n-2} (S_{j+1} - S_j) e^{-i(j+1)\xi}. \tag{11.1}$$

Consequently, for $0 < |\xi| \le \pi$,

$$|\widehat{S}(\xi)| \le \frac{S_1 + S_{n-1} + \mathrm{TV}(S)}{2|\sin(\xi/2)|}.$$

The zero mode is $\widehat{S}(0) = \binom{n}{2}$. The apparent singular denominator at zero is not an error: the estimate is simply not asserted there.

For $L \ge n$, zero-pad S to length L and take

$$\widehat{S}_\ell = \sum_{j=0}^{L-1} S_j e^{-2\pi i \ell j / L}, \qquad S_j = \frac{1}{L} \sum_{\ell=0}^{L-1} \widehat{S}_\ell e^{2\pi i \ell j / L}.$$

The geometric-series identity makes this exactly invertible. An FFT accelerates this coordinate change after the counts are known; it does not compute the incidence geometry for free.

Direct corpus caution. TR9's standard radix-2 discussion contains a normalization/sign slip: $W_{L/2}^{-\ell L/2} = 1$, not $(-1)^\ell$. For unnormalized half-sums the outside multiplier is $1/L$, not $1/2$. The formulas above are derived directly and do not depend on that passage or on a general complexity lower-bound claim.

No exponent from inversion. The concentrated nonnegative vector discussed in Section 4 has an entirely legitimate Fourier transform. Changing coordinates does not exclude it. A useful Ram certificate must add realizability-sensitive bounds on weighted mass, variation, or frequency bands.

12 A Path-Graph Ram Reformulation

For $m = |W|$, arrange the window coordinates into $\boldsymbol{s} = (S_a, \dots, S_b)^T$. Let D be the $(m-1) \times m$ first-difference matrix and $L = D^T D$ the path-graph Laplacian. Then

$$\boldsymbol{s}^T L \boldsymbol{s} = \sum_{j=a}^{b-1} \Delta_j^2.$$

The exact signed identity has the form

$$S_k - \frac{1}{m} \mathbf{1}^T \boldsymbol{s} = \boldsymbol{c}^T D \boldsymbol{s},$$

where $c_j = (j - a + 1)/m$ for $j < k$, and $c_j = -(b - j)/m$ for $j \geq k$.

Proposition 12.1 (Energy certificate)**.** *For $m \geq 2$,*

$$\left| S_k - \frac{1}{m} \sum_{i=a}^{b} S_i \right| \leq A_{m,k} \sqrt{\boldsymbol{s}^T L \boldsymbol{s}}, \tag{12.1}$$

$$A_{m,k}^2 = \frac{1}{m^2} \left(\sum_{t=1}^{k-a} t^2 + \sum_{t=1}^{b-k} t^2 \right). \tag{12.2}$$

The coefficient is sharp over real sequences and also over nonnegative sequences.

Proof. Apply Cauchy–Schwarz to $\boldsymbol{c}^T D \boldsymbol{s}$. Every vector of differences is attained by a sequence, since D is onto. Choose $D\boldsymbol{s} = \boldsymbol{c}$ for equality. Adding a sufficiently large constant to every coordinate preserves both sides and makes the sequence nonnegative. □

For a symmetric radius-r window this coefficient is

$$A_{2r+1,k}^2 = \frac{r(r+1)}{3(2r+1)}.$$

Thus an energy bound can replace a total-variation bound when it scales strongly enough. The coefficient grows like $\sqrt{r/6}$; an unqualified claim that smoothing automatically improves concentration would miss this factor.

In the spectral notation of TR190, choose an orthonormal eigenbasis $L = U \Lambda U^T$. The energy is $\sum_\ell \lambda_\ell |(U^T \boldsymbol{s})_\ell|^2$. The zero eigenvector is constant, and nonzero modes measure deviation from the mean. This is a literal finite graph specialization. It does not require a claimed inverse for arbitrary coarsening. If $m = 1$, the residual and energy are both zero and the statement reduces to equality.

13 Finite Crossing Inputs

The source's finite section reduces activity to endpoint exchanges. For activity threshold $b \geq 4$, it shifts rank by $c = \lfloor b/4 \rfloor$ in one of two classes, obtaining endpoint distance at least c. It then forms greedy batches with endpoint distance $a = \lceil c/2 \rceil$. The inequalities $b/16 \leq a \leq b/4$ preserve scale up to absolute constants. For $b < 4$, strict growth of (4.2) supplies endpoint distance at least one.

Each distance-a batch contributes the a^2 segments between lost and gained points. Every segment has its slope in that batch's open slope interval, so distinct batches cannot contribute the same pair. The source partitions these edges into at most aS convex chains and aS concave chains when S endpoints participate in a window. A convex and concave chain have at most two proper transverse intersections, giving $\mathrm{cr} \leq 2a^2S^2$. Here cr counts crossing pairs in the drawing, not the minimum crossing number over drawings.

Lemma 13.1 (Designated-endpoint crossing bound)**.** *For a simple straight-line drawing on v vertices, with no vertex in an edge interior and no crossings of incident edges, suppose every edge meets a designated set of at most $m \leq v$ vertices. Then*

$$e^3 \leq 2304mv \ \mathrm{cr} + 13824v^3.$$

If $e \geq 24v$, the last term may be omitted.

Proof. If $e = 0$ the assertion is immediate. Otherwise, in the dense case, randomly put each designated vertex on a first side with probability $1/2$, and all others on the second side. Each edge crosses this cut with probability $1/2$, including an edge with two designated endpoints. A bipartite subgraph therefore has $e_0 \geq e/2$ edges and side sizes at most m, v. Sample its sides with probabilities $p = 12v/e_0$ and $q = 12m/e_0$. These are at most one. Deleting one edge per crossing gives $e' \leq 3v' + \mathrm{cr}'$. Expectations yield

$$pqe_0 \leq 3(pm + qv) + p^2q^2 \,\mathrm{cr}\,.$$

Crossing edges have four distinct endpoints, justifying p^2q^2. Substitution gives $e_0^3 \leq 288mv\,\mathrm{cr}$, hence the dense result. If $e < 24v$, then $e^3 < 13824v^3$. □

Combining this with $e = Ra^2$ and $\mathrm{cr} \leq 2a^2N^2$ recovers $R = O((N/a)^{4/3})$, including the sparse alternative. The lemma and its source constants are rederived here; the geometric upgrade from big-O to uniform little-o remains separate.

14 Finite Packing Interface

The independent audit [13], formulas A1–A3, checks the source recurrence conditional on its packing premise and sharpens its finite interface. These refinements do not establish that premise. Let $F(n)$ be the all-ranks maximum, $F(0) = F(1) = 0$, and $G(t) = \max_{0 \le u \le t} F(u)$. Let $K(N, b)$ be the maximum number of step-disjoint activity-at-least-b intervals over all generic N-point sets and all ranks.

Proposition 14.1 (Exact finite recursion)**.** *For integers $2 \le d < N$,*

$$F(N) \le K(N, d-1)G(d) + G(d-2).$$

Proof. Partition at activity threshold $d-1$. Each completed block has at most d active points, and the remainder has at most $d-2$. Apply the exact restriction lemma. G is used deliberately: monotonicity of F itself is neither asserted nor required. □

Theorem 14.2 (One-scale criterion)**.** *Fix integers $q \ge 2$, $N_0 \ge 2q$, $B \ge 2$. Suppose, uniformly for every $N \ge N_0$,*

$$K(N, \lfloor N/q \rfloor - 1) \le B - 1.$$

For every $\beta > 0$ with $B \le q^\beta$,

$$F(N) \le C_\beta N^\beta, \quad C_\beta = \max\left(1, \max_{1 \le n < N_0} \frac{\binom{n}{2}}{n^\beta}\right).$$

Proof. The greedy partition has at most B blocks of active size at most $\lfloor N/q \rfloor < N$. The base cases follow by definition of C_β. Strong induction on the actual active sizes gives $F(N) \le C_\beta B (N/q)^\beta \le C_\beta N^\beta$. Equality $B = q^\beta$ causes no difficulty. □

Only one uniform scale is needed for this implication. The source's assertion for every small packing constant is a sufficient, stronger premise. A bound for one configuration or only middle rank is not the displayed hypothesis. The proposition can also serve as a finite certificate interface for exact small-size packing computations, but no extrapolation beyond their proved range is permitted.

15 Fixed-Scale Recurrence With Constants

Hypothesis 15.1 (Uniform activity packing)**.** For every $\eta > 0$ there is $R_\eta < \infty$ such that, for every generic N-point configuration, every rank, and every integer $1 \le b \le N$ satisfying $N/b \ge R_\eta$, any family of step-disjoint intervals of activity at least b has cardinality at most $\eta(N/b)^{4/3}$.

Theorem 15.2 (Conditional fixed-scale transfer)**.** *Set $a_0 = 4/3$, $\eta = (4 \cdot 2^{a_0})^{-1}$ and choose once an integer $q \ge \max(16, R_\eta)$. Define*

$$\beta_0 = \frac{4}{3} - \frac{\log 2}{\log q}, \qquad C_0 = \frac{1}{2}(4q)^{2-\beta_0}.$$

Under Hypothesis 15.1, every rank sweep on $N \ge 1$ points satisfies $S_k(P) \le C_0 N^{\beta_0}$.

Proof. Let $F(n)$ be the maximal switch count over exactly n generic points and all ranks, with $F(0) = F(1) = 0$. A finite maximum exists: counts are integers in $[0, \binom{n}{2}]$. For $N \ge 4q$, put $d = \lfloor N/q \rfloor$ and $b = d - 1$. Since $N/q \ge 4$,

$$N/(2q) \le b < N/q, \qquad q < N/b \le 2q.$$

Greedy activity blocks have at most $b + 1 = d \le N/q$ active points, by Lemma 6.2. The hypothesis bounds completed blocks by $\eta(2q)^{a_0} = q^{a_0}/4$. One remainder gives at most $q^{a_0}/4 + 1 \le q^{a_0}/2$ blocks.

For each block of active size t_i, Lemma 6.1 gives at most $F(t_i)$ switches at its own fixed local rank. For $1 \le n < 4q$, $F(n) \le n^2/2 \le C_0 n^{\beta_0}$. Strong induction now gives

$$F(N) \le \sum_i F(t_i) \le C_0 \frac{q^{a_0}}{2} (N/q)^{\beta_0} = C_0 N^{\beta_0},$$

because $q^{a_0 - \beta_0} = 2$. Empty blocks contribute zero. □

The source chooses the smaller saving $\log 2/(2 \log q)$, leaving contraction factor $1/\sqrt{2}$. For this homogeneous recurrence no additive toll is present, so the endpoint equality above is admissible. This improves only the conditional numerical extraction from a supplied valid q. No numerical q is supplied by this report or by the source.

16 Integer Branching and Recurrence Robustness

Integer block counting can preserve more of the same argument. Increase the fixed scale, if necessary, to $q \geq 64$ and put

$$B_q = \left\lfloor q^{4/3}/4 \right\rfloor + 1, \qquad \beta_q = \frac{\log B_q}{\log q}, \qquad C_q = \frac{1}{2}(4q)^{2-\beta_q}. \tag{16.1}$$

For $q \geq 64$, $q < B_q < q^{4/3}$, so $1 < \beta_q < 4/3$.

Corollary 16.1 (Integer-branch transfer)**.** *If the same packing hypothesis is valid at this q, then $F(N) \leq C_q N^{\beta_q}$ for every $N \geq 1$.*

Proof. The number of completed blocks is an integer at most $q^{4/3}/4$. Including the remainder gives at most B_q blocks. Repeat the preceding induction with $B_q q^{-\beta_q} = 1$. The base bound is unchanged in form. For $q > 64$, $q^{4/3}/4 > q$, and its floor plus one is larger than q; at $q = 64$, $B_q = 65 > 64$. The other inequalities are immediate. □

For illustration only, at a hypothetical valid $q = 1024$, $B_q = 2581$ and $\beta_q = \log_{1024} 2581$. This is not a certified scale for halving lines. The necessary lower bound below gives no upper bound on the geometric threshold.

Why endpoint induction needs an exact recurrence. Suppose instead an implementation only proves

$$F(N) \leq BF(\lfloor N/q \rfloor) + DN^{\gamma}, \qquad D > 0.$$

At $\beta = \log_q B$, the homogeneous term uses the entire induction budget. One cannot absorb the toll by merely increasing the same constant. For $\beta > \max(\log_q B, \gamma)$,

$$Bq^{-\beta} < 1, \qquad C \geq \frac{D}{1 - Bq^{-\beta}}$$

(together with a base-case bound) suffices for $N \geq 1$. This follows by inserting Cn^{β} and using $N^{\gamma} \leq N^{\beta}$.

The present geometric restriction recurrence has no toll because every switch is assigned to exactly one block and then to the exact restricted sweep. A future approximate Ram compression must either prove that same exactness or retain its residual toll. This is a concrete connection to the error discipline of TR165, not a blanket assertion that all residual decompositions contract.

17 Necessary Scale and Perturbation Sanity Checks

Proposition 17.1 (Necessary scale)**.** *For $\eta = 2^{-10/3}$, any integer $q \geq R_\eta$ valid under Hypothesis 15.1 must satisfy $q \geq 1024$.*

Proof. Take N points on a strictly convex lower chain, with distinct first coordinates. A sufficiently small generic perturbation preserves the property that every point appears on the lower envelope of $y - sx$. Thus the rank-one sweep has $N - 1$ switches, visiting points successively without return. A consecutive block of $b - 1$ switches has exactly b active points.

Let $N = qd$, $b = d - 1$, and let d tend to infinity through integers at least four. At least q disjoint blocks of $d - 2$ switches fit into the $qd - 1$ switches. Since $N/b > q \geq R_\eta$, packing implies

$$q \leq \eta \left(\frac{qd}{d-1} \right)^{4/3}.$$

Letting $d \to \infty$ gives $q \leq \eta q^{4/3}$, hence $q \geq \eta^{-3} = 1024$. □

This is a necessary condition only. It rules out promoting a formally small q in recurrence algebra to an actual geometric threshold. It supplies neither $R_\eta \leq 1024$ nor an explicit power saving.

Sharper finite ratio. For $N \geq 4q$, $b = \lfloor N/q \rfloor - 1$ satisfies

$$\frac{N}{b} \leq \frac{5q - 1}{3}.$$

Indeed, writing $d = \lfloor N/q \rfloor \geq 4$ gives $N/b \leq [q(d+1) - 1]/(d-1) = q + (2q-1)/(d-1)$. The maximum occurs at $d = 4$. Thus one may replace B_q by $\lfloor \eta((5q-1)/3)^{4/3} \rfloor + 1$ in the conditional one-scale criterion whenever its hypotheses hold. Optimality among all decompositions is not asserted.

Projection-order margins. Order-type preservation alone does not fix a chosen starting-angle half-turn ledger; its strict initial projection ordering must also be preserved. For dual-level counts preserve first-coordinate order as well: the height sign at a pair event is the sign of $\det(p_j - p_i, p_\ell - p_i)/(x_j - x_i)$. This records the numerator and denominator conventions needed for exact perturbation transfer.

18 Audit of Rank Coordinates and Measures

The source argues by contradiction from consecutive distance-a batches with $N/a \to \infty$ and $R \asymp (N/a)^{4/3}$. Its normalizations are

$$E_* = Ra^2, \qquad M = aN, \qquad E_*^3 \asymp a^2 N^4.$$

These are three different object counts. Vertices receive weight N^{-1}, edges E_*^{-1}, and crossings M^{-2}. Confusing these normalizations invalidates the absolute-continuity step even if a formal transform is invertible.

For a dyadic time window J, the source extends convex chains to graphs and smooths their height ranks:

$$r_J(x, y) = \frac{1}{M} \sum_{i=1}^{M} \Phi\big((y - g_{J,i}(x))/\epsilon_J\big).$$

Separated slope windows $A < B$ give intersecting level graphs and a homeomorphism $(r_A, r_B) : \mathbb{R}^2 \to (0, 1)^2$. The rank-slot estimate at crossings gives a planar projection bound for the crossing measure σ_{AB}, not initially for the vertex measure μ.

The edge measure ν records an endpoint label and batch time; its endpoint marginal is λ, with window marginal λ_J. The distinct participating-vertex measure is μ_J, and $s_J = \mu_J(Z)$. The localized crossing calculation divides

$$e(U)^3 \le 4608 a^2 m(U) S_J^3 + 13824 S_J^3$$

by E_*^3. The sparse term is at most a constant times $1/(a^2 N)$ and tends to zero. Neighborhood approximation under weak convergence then gives the source's

$$\lambda_J(F)^3 \le C s_J^3 \mu_J(F), \qquad \lambda \ll \mu. \tag{18.1}$$

The source further proves nonatomic conditional time distributions and absolute continuity of positive edge mass with respect to a countable mixture of crossing measures in its time window. The tail bound is $\#\{J : s_J > u\} \le C u^{-4/3}$ at a fixed resolution. These properties are used to select transverse directions, not to identify the spectrum with an arbitrary graph signal.

Audit boundary. The finite algebra and vanishing sparse term have been checked. The weak-limit and selection statements are read and traced as source dependencies. This report does not label the whole chain an independent verification. In particular, batch participants omit points that change and return *inside* one batch; the earlier activity reduction is essential for covering the original sweep.

19 Differentiability and the Zoom Quantifiers

In `differentiability.tex`, coordinates from other separated windows become coordinatewise monotone functions f_C of a chosen pair-coordinate point $p = (z_A, z_B)$. The source supplies a monotone differentiability argument, an equal-dimensional critical-image lemma, and a selection of independent gradients. The critical-image step concerns *all* singular differentiability points, not only almost all points for the domain Lebesgue measure: crossing measures may be singular in that domain chart.

From the signs of directional derivatives it defines allowed direction intervals $D_p(t)$. Nonatomic conditional time mass and crossing absolute continuity produce three distinct projective directions. The choices initially depend on p. Countable selection also supplies a fixed triple of times for a common normalization on positive edge mass.

The lemma in `zoom.tex` is stronger than an assertion that an already existing limit is affine. It must prevent collapse before a limit map exists. It assumes:

1. homeomorphisms onto domains exhausting the plane;
2. fixed signed monotonicity after one target linear change;
3. control of every bounded limiting displacement on each of three line families, even with moving base points;
4. a fixed origin and a unit normalization.

The source proof uses pointed cones, parameterized entire-line images, transverse intersections, rational grids, and finally inverse convergence. Merely citing exact affine rigidity would not supply these sequential hypotheses.

Its radial conclusion is, for each good p, some $C_p < \infty$ such that for every integer $T \geq 2$, all sufficiently small *fixed* $h > 0$ satisfy eventually in configuration index n,

$$V_n(p, h/T) \geq \frac{U_n(p,h)}{C_p T}, \qquad U_n(p, h/T) \leq \frac{C_p V_n(p,h)}{T}. \tag{19.1}$$

The radius bound may depend on p, T and the eventual index on p, T, h. No uniform convergence rate is obtained.

This is a genuine analytical burden in the source claim. A finite Ram reconstruction, a coordinate homeomorphism, or numerical nondegeneracy at sampled points does not establish (19.1) on the positive-measure set required later. We preserve this burden explicitly rather than restating it as a newly proved Ram theorem.

20 Spatial Selection and Dimension

The source's spatial section carefully separates edge and vertex density. Its selected edge mass has absolutely continuous projection in the rank square. This alone does not make the entire vertex projection absolutely continuous. It decomposes

$$\pi_{\#}\mu = f(p)\,\mathrm{d}p + \sigma_s,$$

deletes the null carrier of σ_s, and restricts to $f \le M$ on a set G retaining $\lambda_{K_0}(G) > 0$. It thereby obtains $\mu(G \cap \pi^{-1}(U)) \le M|U|$.

For $\eta = 1/16$, countable selection fixes T, m and radii $h_j = T^{-m-j}$ satisfying eventual radial estimates

$$V_n(p, h_{j+1}) \ge T^{-1-\eta} U_n(p, h_j), \qquad U_n(p, h_{j+1}) \le T^{-1+\eta} V_n(p, h_j).$$

The index thresholds still depend on p, j. A single Hausdorff limit of compactified graphs is used instead of an unjustified uniform diagonal over uncountably many centers.

The source claims a single-valued graph l on the selected rank points, satisfying

$$|l(p) - l(q)| \ge c_G |p - q|^{1+\eta}. \tag{20.1}$$

Its proof fixes a center and a radius before taking the configuration limit, and only then shrinks the radius. The restriction to G occurs after weak convergence, not by assuming convergence of arbitrary finite Borel restrictions.

Proposition 20.1 (Conditional dimension calculation)**.** *If bounded rank density and* (20.1) *hold and the position measure is supported in a bounded disk, then*

$$\nu_G(B(x, r)) \le C_G r^{2/(1+\eta)}, \qquad \nu_G(\textit{strip of width } r) \le C'_G r^{2/(1+\eta)-1}.$$

Proof. Any two rank points over a radius-r physical disk are at distance at most $(2r/c_G)^{1/(1+\eta)}$. Their set has outer area at most a constant times its diameter squared. Bounded rank density proves the disk estimate. In a bounded support, a strip is covered by $O(1/r)$ radius-$O(r)$ disks. Sum the disk estimates. □

At $\eta = 1/16$, the exponents are $32/17$ and $15/17$. The calculation is verified here conditional on its inputs. Obtaining those inputs from every hypothetical extremal sequence is part of the source's still-unadopted global geometry.

21 Sharp Finite Three-Strip Bound

We now prove a local strengthening of the source's boundary lemma. Let P be finite, $B \subset \mathbb{R}^2$ bounded and open, and $D \geq \max(1, \sup_{x \in B} |x|)$. Suppose equal-cardinality marked sets are strictly cut out by

$$M_i = \{x \in P : \nu_i \cdot x < c_i\}, \quad |\nu_i| = 1, \quad |\nu_i - \nu_0| \leq \theta, \quad 0 \leq i \leq m.$$

Choose any matching of the lost to gained points at each step. Let C_B count transfers with exactly one endpoint in B. Put $\rho = D\theta$, $c_- = \min_i c_i$, $c_+ = \max_i c_i$, and

$$A = \bigcup_{i=1}^{m} (M_{i-1} \triangle M_i), \qquad S = \bigcup_{q \in \{c_0, c_-, c_+\}} \{x : |\nu_0 \cdot x - q| \leq 3\rho\}.$$

Theorem 21.1 (Boundary coefficient one)**.** *Under these assumptions,*

$$\#((A \cap B) \setminus S) \leq C_B, \qquad \#(A \cap B) \leq C_B + \sum_{q \in \{c_0, c_-, c_+\}} \#\{x \in P \cap B : |\nu_0 \cdot x - q| \leq 3\rho\}. \quad (21.1)$$

Proof. Write $t(x) = \nu_0 \cdot x$ and $e_i(x) = (\nu_i - \nu_0) \cdot x$, so $|e_i(x)| \leq \rho$ in B. If $x \in (A \cap B) \setminus S$ is initially marked, then $t(x) < c_0 - 3\rho$. It is unmarked at some state, whence $t(x) > c_- - \rho$. Avoiding the strip at c_- strengthens this to $t(x) > c_- + 3\rho$. Thus x is absent at a state i_- attaining c_-.

If this class is nonempty, $c_0 - c_- > 6\rho$. Every initially unmarked $y \in P \cap B$ then remains unmarked at i_-: $t(y) + e_{i_-}(y) > c_0 - \rho > c_-$. Thus all inside-$B$ membership differences from state 0 to i_- are losses. Their number equals token exits minus token entries during that prefix, and is at most the total number E_B of exits during the entire sequence. This bounds the first class.

For an initially unmarked participant outside S, the reversed argument gives $c_0 + 3\rho < t(x) < c_+ - 3\rho$, so it is present at a state i_+ attaining c_+. If this class is nonempty, $c_+ - c_0 > 6\rho$, and every initially marked point of B remains marked at i_+. The net change inside B is all gains, at most the total number I_B of entries. The two disjoint classes therefore total at most $E_B + I_B = C_B$. Adding strip participants proves the result. □

22 Sharpness and Token Budgets

The improvement in Theorem 21.1 is specifically over the displayed $2C_B$ in the source. Its proof bounded each class separately by all transfers. Keeping exits and entries separate removes this duplication. No participant-by-participant assignment to its own token is required.

Proposition 22.1 (Sharp example)**.** *Coefficient one in* (21.1) *cannot be decreased under the theorem's stated assumptions.*

Proof. Take $P = \{(0,0), (-20,4)\}$ and let B be the open unit disk, $D = 1$. Choose

$$\nu_0 = (0,1), \quad c_0 = 3, \qquad \nu_1 = (3/5, 4/5), \quad c_1 = -3.$$

Initially only $(0,0)$ is marked; finally only $(-20,4)$ is marked, because its second projection is $-44/5 < -3$. There is one matching transfer, an exit from B, so $C_B = 1$. Here $\theta = |\nu_1 - \nu_0| = \sqrt{2/5}$ and all three strips have centers among $-3, 3$. The point $(0,0)$ has distance 3 in the t-coordinate from either center, exceeding $3\sqrt{2/5}$. Thus exactly one participant of B is outside all strips. Equality holds. □

The example establishes sharpness for the stated finite cut lemma, not for every more constrained extremal halving construction. The proof includes $\theta = 0$, escaping outside points, returning memberships, repeated thresholds, and arbitrary matchings. Equal cardinalities ensure every lost point has a gained partner.

For the source application, transfers over a selected window have directions in a pointed cone. Choose two independent linear functionals positive on that cone, and a parallelogram bounded by two levels of each functional. Every token's two functional values are monotone. Each token crosses each of the four side levels at most once. With at most N tokens, the total number of transfers having exactly one endpoint in the parallelogram is at most $4N$.

After normalization and fixed-resolution limits, the source obtains budgets $\sum_J b_J \le 4$ and $\sum_J \theta_J \le \pi$. Replacing its finite $2C_B$ bound by (21.1) changes the limiting estimate to

$$\mu_J(G) \le b_J + C_G \theta_J^{d-1}.$$

The same continuous-test argument off the limiting strips justifies this local substitution. It changes a coefficient, not the dimension threshold or the final finite exponent.

23 A Conditional Boundary Contradiction

The remaining calculation can be abstracted without claiming the source's spatial construction. At each dyadic mesh δ, suppose there are at most δ^{-1} windows and nonnegative numbers $\lambda_J(G), s_J, \mu_J(G), b_J, \theta_J$ satisfying

$$\lambda_J(G) \le \delta, \qquad \lambda_J(G)^3 \le C s_J^3 \mu_J(G), \tag{23.1}$$

$$\#\{J : s_J > u\} \le A u^{-4/3}, \qquad \sum_J b_J \le B, \quad \sum_J \theta_J \le \Theta, \tag{23.2}$$

$$\mu_J(G) \le b_J + K\theta_J^{\alpha}, \qquad 3/4 < \alpha < 1. \tag{23.3}$$

Assume the total $L = \sum_J \lambda_J(G)$ is independent of mesh. All constants here are independent of δ.

Theorem 23.1 (Abstract vanishing criterion)**.** *Under these hypotheses, $L = 0$. More precisely, for $M \ge 1$,*

$$L \le A M^{-4/3} + C^{1/3} M \left(B^{1/3}\delta^{1/12} + K^{1/3}\Theta^{\alpha/3}\delta^{\alpha/3 - 1/4} \right). \tag{23.4}$$

Proof. Windows with $s_J > M\delta^{3/4}$ number at most $AM^{-4/3}\delta^{-1}$ and each contribute at most δ. For the others, the cubic coupling gives

$$\sum_J \lambda_J(G) \le C^{1/3} M \delta^{3/4} \sum_J \mu_J(G)^{1/3}.$$

Hölder bounds the last sum by $\delta^{-2/3}(\sum_J \mu_J(G))^{1/3}$. Concavity yields $\sum_J \theta_J^{\alpha} \le \delta^{\alpha-1}\Theta^{\alpha}$. Combine the inequalities and use $(x+y)^{1/3} \le x^{1/3} + y^{1/3}$. Both powers of δ are positive. First let $\delta \downarrow 0$ at fixed M, then $M \to \infty$. □

For the source's $\alpha = 15/17$, the slow power is $3/68$. Even choosing $M = \delta^{-9/476}$ in the already limiting inequality gives $L \le C'\delta^{3/119}$ for $0 < \delta \le 1$. This optimized rate is solely a bound for an abstract limiting mass; it is *not* ε in (1.1). No convergence rate back to finite N has been supplied. The threshold $d = \alpha + 1 > 7/4$ follows exactly from $\alpha/3 - 1/4 > 0$; changing a boundary coefficient cannot lower it.

24 Rank-Sensitive Consequences and Cutlines

Suppose, conditionally, that every complete straight-line arrangement of m lines has each level of complexity at most Am^{β}, uniformly in the level, where $1 < \beta < 4/3$. Fix constants C_s, C_c, C_b for a planar shallow cutting: there are at most $C_s n/(k+1)$ cells, at most $C_c(k+1)$ conflict lines per cell, and at most $C_b(k+1)$ boundary contributions per cell. Then

$$\operatorname{comp}(\Lambda_k) \le C_s\big(AC_c^{\beta} + C_b\big)n(k+1)^{\beta-1}. \tag{24.1}$$

Indeed, inside each cell nonconflicting lines contribute a fixed offset to the rank. Apply the all-ranks estimate to that cell's conflict list and add its boundary cost. Since $\beta > 1$, the linear boundary term is absorbed by $(k+1)^{\beta}$.

This proves the algebraic transfer with explicit supplied cutting constants; it does not produce numerical constants for the cited shallow-cutting theorem. Source `recurrence.tex` cites its qualitative form through [7], Lemma 3.2.

Cutline conventions. Under $p = (x, y) \mapsto \{(s, t) : t = y - sx\}$, a dual vertex with exactly j strict lines below is a rank-$(j+1)$ switch. A level Λ_j has vertices from both V_{j-1} and V_j. Ignoring one set creates an off-by-one counting error. At boundaries of a simplex, conflict lines meet sides at most twice; complete straight lines and general position are part of the assumptions. This argument is not automatically valid for pseudolines, segments, or arbitrary sampled graph drawings.

Two independent conditional routes. Hypothesis 10.1 gives k-sensitive bounds directly through the spectrum. Hypothesis 15.1 first gives an all-ranks bound through exact block restriction; a shallow cutting then restores k-dependence. These hypotheses are not asserted equivalent. Their common unresolved feature is a uniform estimate that uses realizability beyond the global swap ledger.

The source's recurrence and the source's compactness proof must therefore be audited separately. Correct recurrence arithmetic cannot repair a failure in the geometric premise, while a local variation coefficient improvement is valid even if that premise is not established.

25 Deterministic Checks and Evidence

The companion `check_m334.py` is a local, deterministic test program using only integer and rational arithmetic for its assertions. It has no networking code and does not alter inputs. It records the interpreter, seed, script hash, assertion counts, exact coordinates, rational event slopes, ranks, and spectra. The final local run passed 209,669 assertions, including 14 input-preservation checks, 21,324 joint-bound checks, 21,324 energy checks, and 13,182 joint-bound attainment cases. The independent audit [13] was read in full after the first draft. Its reported 1,443,762 assertions are separate evidence, not included in this report's own check totals; its geometric proof-certification boundary is retained.

Algebraic suite. For every ternary sequence of lengths one through seven and every target coordinate, it checks signed telescoping and the sharp variation inequality. Separate step examples attain the coefficient. Positive rational weights check the weighted identity over lengths one through seventeen. Cube values of q make $q^{4/3}$ exact for recurrence floor tests. The identities $15/51 - 1/4 = 3/68$ and $32/17 > 7/4$ are rationally checked.

Geometric suite. With fixed seed 334, eight sweep-generic rational-coordinate examples are constructed for every $n = 2, \ldots, 12$, giving 88 configurations. All event slopes are required to be distinct. The extra source condition excluding triple proper segment crossings is not certified by this generator and is not needed for the tested identities. Exact event sorting checks adjacency, reversed terminal order, the global swap budget, full-turn counts against determinant side counts, and middle-rank halving counts for even sizes. Selected consecutive state intervals test active-subset rank restriction. Every threshold from one to n tests greedy overshoot and the unfinished remainder.

Boundary witness. The two-point example in Section 22 is checked using exact cuts and squared distances. No floating approximation to its unit normals or strip separation is used. The symbolic proof, not the example alone, establishes coefficient one in general.

Evidence files. `evidence/check-results.json` contains the exact assertion totals and pass status. `evidence/sweep-witnesses.json` retains the geometry and ledgers. `evidence/input-hashes.json` records the complete read scope and original hashes. The final `evidence/final-manifest.json` binds the deliverables and audit outputs with SHA256 hashes and lists all changed paths.

What has not been computed. The examples are neither an exhaustive enumeration of order types nor a proof of asymptotic packing. We do not certify extremal $h(n)$, stretchability of arbitrary allowable sequences, the source's compactness subsequences, any value of R_η, or a global power-saving exponent. A passing finite suite is evidence for the local implementation, not a universal incidence theorem.

26 Pros, Limitations, and Future Work

Advantages. The Ram ledger isolates exact count identities from geometric inequalities. Weighted residuals make each loss visible and permit rigorous optimization once bounds are available. Exact restriction preserves local ranks, while the path-graph energy formulation offers an alternative, quantitatively normalized certificate. The sharp boundary constant and integer recurrence extraction provide concrete local improvements with short, auditable proofs.

Limitations. Fourier inversion and graph functional calculus are changes of coordinates. They cannot eliminate nonrealizable concentration patterns without a realizability theorem. Sharpness among nonnegative vectors is not sharpness among point configurations. Finite coordinate tests sample a small family of realizable examples and say nothing universal about large n. The source's spatial construction remains the main dependency of its claimed power saving; it is not independently validated by the corpus comparison.

Focused research tasks. First, derive realizability-sensitive estimates for the one-sided signed residual in (7.2), rather than immediately discarding cancellation. Second, investigate energy estimates strong enough to offset the explicit $\sqrt{r}$ window factor in Section 12. Third, test whether degree constraints on halving graphs can improve these bounds without assuming arbitrary degree lists are geometrically realizable.

Fourth, formalize the active-subset restriction and block partition before attempting the measure-theoretic part. Fifth, audit the source's moving-base-point hypotheses, fixed-radius quantifiers, graph-fiber uniqueness, and positive edge-mass retention as separate lemmas. Sixth, seek an effective modulus for the claimed uniform packing estimate. Only such a modulus would turn a conditional q-formula into numerical halving-line constants.

Priority distinction. Testing more examples is useful for finding mistakes or candidate inequalities. It is not the most direct way to establish an uncountable weak-limit selection or a uniform large-size bound. Likewise, reducing the coefficient of C_B from two to one does not address the dimensional threshold $7/4$. Effort should be allocated according to which premise a proposed result actually strengthens.

Conclusion. The defensible progress is local: exact counting, sharp constants, conditional transfer, and reproducible certificates. The report does not claim to solve the global halving-line problem or to independently verify the pinned preprint's new geometric theorem.

27 Provenance and Reproduction

The workspace root is

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math`

and the source build directory, relative to it, is

`_sources/openai-math/preprints/`
`A-power-saving-for-planar-halving-lines-September-25-2026/build`

Complete public-source read scope: `main.tex` and `references.bib`. Within `sections/`, all of `introduction.tex`, `finite.tex`, `rank-limits.tex`, `differentiability.tex`, `zoom.tex`, `spatial.tex`, `boundary.tex`, and `recurrence.tex` were read completely. Only row 334 of the two structured catalogue files was needed for the mathematical assessment; no other manuscript was audited.

The principal private source is

`C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/`
`res-math/TR-item-214-astra.tex`.

It was read in full, not merely its routing entry. The selected TR9, TR165, and TR190 passages and their exact absolute paths are listed in the evidence inventory. Neither the entire TR1–250 corpus nor the complete 722-manuscript collection was read or certified.

Build procedure. The native editor and native standalone compiler were attempted with bounded waits of 15 and 30 seconds, respectively. Both attempts timed out unconfirmed; native compilation success was not established. These outcomes are recorded separately from the successful first MiKTeX on-disk export and the final export diagnostics. The existing helper is invoked with the bundled Python:

`_tools/compile_report.py`
`M334-Planar-Halving-Lines/TR-M334-Ram-Halving.tex`
`-round 1`, followed by `-round 2`.

The helper uses the installed MiKTeX and Poppler; nothing is installed. At most two formatting audit rounds are used, with up to three TeX passes in each. All pages are rendered for visual inspection, with detailed page review where formulas or tables need it.

Freeze. The final manifest records output hashes, input preservation, build status, page count, changed-path inventory, and visual audit scope. It does not include itself in its own hash list. The manifest is a reproducibility record, not a proof certificate. The final source, PDF, check program, and evidence are left frozen after this record is written.

References

[1] OpenAI. *A power saving for planar halving lines*. Preprint, September 25, 2026. Public pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-power-saving-for-planar-halving-lines-September-25-2026/main.pdf`. Complete local LaTeX read; global claim not independently certified here.

[2] OpenAI. Collection README at the same pinned commit. `https://raw.githubusercontent.com/openai/math/adc7f1241b42e322a6451854ab7e4b4c146bf78a/README.md`. Browsed October 7, 2026; collection verification disclaimer.

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[10] M. Subbarao. *FFT Applicability Analysis for the Shift-Variant Ram Transform: Detailed Analysis and Special Cases*. TR9, April 4, 2026, confidential local corpus. Selected lines 1–240; standard-DFT specialization only.

[11] M. Subbarao. *Analytic Ram Core and Learned Residual*. TR165, confidential local corpus. Selected residual/error passages, split nonuniqueness, and limitations. No learning or uniform residual-smallness theorem is imported here.

[12] M. Subbarao. *The Graph-Global Ram Transform (GGRT): A Master Transform on Graphs, with a Restriction Theorem Unifying Multiresolution Graph Transforms*. TR190, June 2026, confidential local corpus. Selected lines 1–260; spectral operator and exact retained-subspace scope used in the path-graph correspondence.

[13] Independent recurrence auditor (Pauli). *M334 Independent Bounded Mathematical Audit*, October 7, 2026. Confidential local artifact `_audit/M334-independent-recurrence-audit.txt`. Read completely and integrated with attribution: A1–A3 finite recurrences, A6 sharp joint mass/variation optimum, necessary $q \geq 1024$, and perturbation/cutline checks. The audit explicitly does not certify the full upstream geometric proof.